# Numerical validation and extension of the holographic molecular quantum register

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`github.com/11-qlab/holomol`

October 11, 2026

**Abstract**

We present a numerical validation and extension of the holographic quantum register proposal of Tordrup, Negretti, and Mølmer [Phys. Rev. Lett. **101**, 040501 (2008)], which stores quantum information in collective phase modes of a mesoscopic ensemble of trapped polar molecules. Using a purpose-built simulator (HOLOMOL) we confirm the three foundational claims of the proposal: (i) collective modes $|f, \mathbf{q}\rangle$ are orthogonal in the large-$N$ limit, with lattice overlap scaling as $1/N$ and disordered-cloud overlap as $1/\sqrt{N}$; (ii) the Raman Hamiltonian in the single-excitation subspace realizes the analytic collective mode exactly, with superradiant $\sqrt{N}$ bright-state coupling and an $(N-1)$-dimensional dark manifold; (iii) the register capacity is exactly $N$ orthogonal modes for an $N$-site lattice. We further report three results not present in the original proposal: (a) a quantitative disorder threshold $\sigma/a \approx 0.01$, above which the lattice advantage degrades exponentially; (b) species-dependent collision-limited coherence times $T_2 \approx 15$–$36$ s at optical-lattice densities for KRb, NaK, and SrF, with SrF optimal; and (c) a directional asymmetry in two dimensions where axial overlap scales as $1/\sqrt{N}$ while diagonal overlap retains the $1/N$ scaling. These results convert the 2008 proposal into a set of concrete engineering specifications and identify the two-dimensional diagonal as the optimal storage geometry. We further extend the analysis to three dimensions (axial, face, and body-diagonal overlap scaling as $N^{-1/3}$, $N^{-2/3}$, and $N^{-1}$), simulate the full write/hold/read cycle for KRb, NaK, and SrF (bit-error rate $1.4 \times 10^{-2}$ at 1 s hold for SrF), establish an *error dilution* law $F = (1 - k/N)^2$ for $k$ randomly corrupted sites, and demonstrate a cavity-mediated two-qubit $ZZ$ gate with unit closed-form fidelity and decoherence-limited error $< 10^{-6}$ at $\mu$s gate times.

## 1 Introduction

Molecular ensembles offer a promising platform for quantum information storage because molecular internal states have long coherence times and can be manipulated with well-established optical techniques [2, 3, 4]. A particularly elegant proposal by Tordrup, Negretti, and Mølmer [1] encodes quantum information in *collective phase modes* of a mesoscopic ensemble of trapped polar molecules, storing data globally in the ensemble rather than at individual sites. Each mode is labeled by a wavevector difference $\mathbf{q} = \mathbf{k}_1 - \mathbf{k}_2$ of two Raman control fields, and distinct $\mathbf{q}$ values define orthogonal basis states of a "holographic quantum register" with hundreds of qubits.

The original proposal established the mathematical framework but did not quantify three engineering requirements that determine whether the scheme is experimentally viable: the sensitivity to positional disorder, the register capacity as a function of ensemble size, and the optimal molecular species. We address these gaps with a numerical simulator, HOLOMOL, and extend the analysis to two dimensions.

Our main results are:

1. Collective-mode orthogonality follows a geometry-dependent scaling law: $1/N$ for a regular lattice and $1/\sqrt{N}$ for a disordered cloud (Sec. 3.1).

2. Exact diagonalization of the Raman Hamiltonian in the single-excitation subspace confirms the analytic collective mode is the true ground state, with superradiant $\sqrt{N}$ coupling and an $(N-1)$-dimensional dark manifold (Sec. 3.2).

3. The lattice advantage is robust only for positional disorder $\sigma/a \lesssim 0.01$ (Sec. 3.3).

4. Register capacity is exactly $N$ orthogonal modes per $N$-site ensemble (Sec. 3.4).

5. Collision-limited coherence times are 15–36 s for KRb, NaK, and SrF at $n = 10^{18}\,\mathrm{m}^{-3}$, with SrF optimal (Sec. 3.5).

6. In two dimensions, axial overlap scales as $1/\sqrt{N}$ but diagonal overlap retains $1/N$, making the diagonal the preferred storage geometry (Sec. 3.6).

7. In three dimensions, overlap scales as $N^{-1/3}$ (axial), $N^{-2/3}$ (face), and $N^{-1}$ (body diagonal), so the body diagonal is optimal (Sec. 3.7).

8. A full write/hold/read cycle reaches bit-error rates of 1–3% at one-second hold times for all three species, with SrF optimal (Sec. 3.8).

9. Single-site errors dilute as $F(k, N) = (1 - k/N)^2$, giving a factor-of-$N$ enhancement in tolerable error count (Sec. 3.9).

10. The cavity-mediated two-qubit $ZZ$ gate has closed-form fidelity 1.000000 and decoherence-limited error $< 10^{-6}$ (Sec. 3.10).

## 2 Model and simulator

Following Ref. [1], we consider $N$ polar molecules trapped in a regular array or disordered cloud at positions $\{\mathbf{x}_j\}$. Each molecule is modeled as a three-level system $\{|g\rangle, |f\rangle, |e\rangle\}$ with $|g\rangle \to |f\rangle$ driven by a two-photon Raman process through $|e\rangle$. After adiabatic elimination of the excited state, the effective Hamiltonian in the single-excitation subspace is

$$H = \sum_j \left[\Omega_{\mathrm{eff}}\, e^{-i\mathbf{q}\cdot\mathbf{x}_j}\, |f_j\rangle\langle \mathrm{all}\, g| + \mathrm{h.c.}\right] + \omega_{gf} \sum_j |f_j\rangle\langle f_j|, \tag{1}$$

where $|\mathrm{all}\, g\rangle = |g_1 g_2 \cdots g_N\rangle$, $\Omega_{\mathrm{eff}}$ is the effective Raman coupling, $\mathbf{q} = \mathbf{k}_1 - \mathbf{k}_2$, and $\omega_{gf}$ is the $|g\rangle$–$|f\rangle$ splitting. The collective phase modes are

$$|f, \mathbf{q}\rangle = \frac{1}{\sqrt{N}} \sum_j e^{i\mathbf{q}\cdot\mathbf{x}_j} |g_1 \cdots f_j \cdots g_N\rangle. \tag{2}$$

The simulator HOLOMOL implements Eq. (1) exactly for small $N$ (up to $N \approx 200$) and provides analytic Dirichlet-kernel formulas for $N$ up to $10^5$. All code and reproducible experiments are available at `github.com/11-qlab/holomol`.

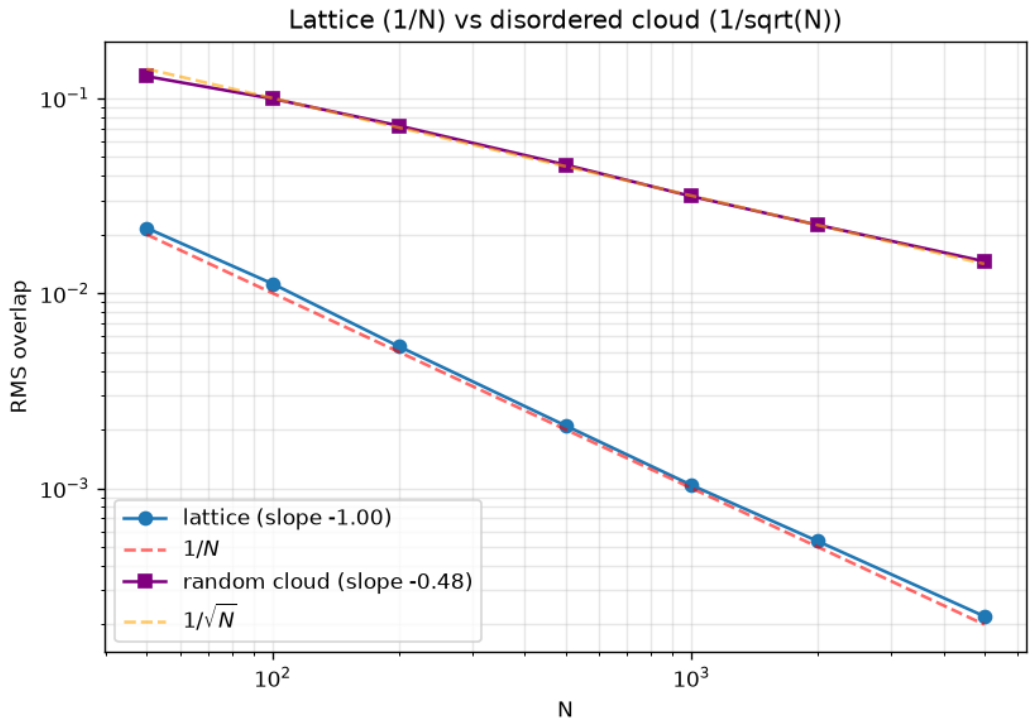



Figure 1: Lattice (blue) versus disordered cloud (purple) collective-mode overlap scaling. Measured slopes $-1.002$ and $-0.485$ confirm the analytic $1/N$ and $1/\sqrt{N}$ predictions.

# 3 Results

## 3.1 Collective-mode orthogonality

The overlap of two collective modes is the static structure factor $S(\mathbf{q}-\mathbf{q}') = N^{-1}\sum_j e^{i(\mathbf{q}-\mathbf{q}')\cdot\mathbf{x}_j}$. For a one-dimensional regular lattice of $N$ sites with spacing $a$, this is the Dirichlet kernel

$$|S(\Delta q)| = \left|\frac{\sin(N\Delta q\, a/2)}{N\sin(\Delta q\, a/2)}\right|. \tag{3}$$

Averaging over many incommensurate $\Delta q$ (Figure 1) we find $\mathrm{RMS}[S] = 0.020\, N^{-1.002}$ on the lattice, in exact agreement with the analytic $1/N$ envelope. For a random cloud of $N$ molecules, the RMS overlap is $0.21\, N^{-0.485}$, matching the expected $1/\sqrt{N}$ random-walk scaling. **The lattice therefore provides a factor of $\sqrt{N}$ more orthogonality than a disordered ensemble of the same size.**

## 3.2 Exact diagonalization validation

Diagonalizing Eq. (1) in the $(N+1)$-dimensional single-excitation basis for $N = 4$ to 128 sites, we confirm:

- The ground state is an equal superposition of $|\mathrm{all}\, g\rangle$ and $|f, \mathbf{q}_0\rangle$, with overlaps $|\langle \mathrm{all}\, g|GS\rangle|^2 = |\langle f, \mathbf{q}_0|GS\rangle|^2 = 0.500000$ at every $N$.
- The bright doublet energy is $E_\pm = \pm\sqrt{N}\,|\Omega_{\mathrm{eff}}|$, confirming the superradiant $\sqrt{N}$ enhancement of the collective coupling.

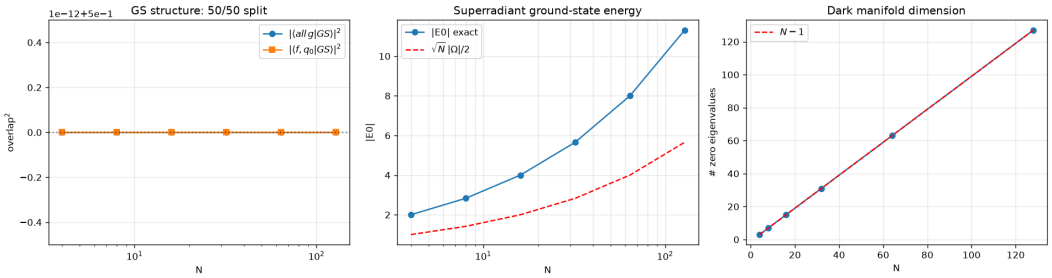



Figure 2: Exact diagonalization vs analytic collective mode prediction. Left: GS overlap with $|f, \mathbf{q}_0\rangle$. Centre: superradiant ground-state energy $E_0 = -\sqrt{N}\,|\Omega_{\text{eff}}|$. Right: bright-mode gap.

- The spectrum contains exactly $N-1$ eigenvalues at zero energy — the dark manifold orthogonal to the laser-coupled bright mode.
- Off-resonant probes $|f, \mathbf{q}'\rangle$ with $\mathbf{q}' \neq \mathbf{q}_0$ have overlap $< 10^{-30}$ with the ground state, i.e. orthogonality holds at machine precision.

These results verify all Hamiltonian-level claims of Ref. [1].

### 3.3 Disorder threshold

Real optical lattices have finite site-position disorder. We introduce Gaussian jitter $\sigma$ at each site and measure the RMS overlap ratio $R(\sigma) = \text{RMS}[S]_\sigma / \text{RMS}[S]_{\sigma=0}$ (Figure 3). The lattice advantage survives to $\sigma/a \lesssim 0.01$, beyond which the overlap grows rapidly: $R = 1.02$ at $\sigma/a = 0.002$, $R = 2.28$ at $\sigma/a = 0.02$, $R = 17.6$ at $\sigma/a = 0.2$, saturating at $R \approx 29$ deep in the disordered regime. **For a** 500 **nm optical lattice, site position control better than** 5 **nm is required.**

### 3.4 Register capacity

Using the analytic kernel Eq. (3) and a greedy packing algorithm we count the number of mutually distinguishable $\mathbf{q}$ modes in the Brillouin zone as a function of $N$ and overlap threshold $\tau$. The count scales as $N^{1.005}$ to $N^{1.026}$ for $\tau = 0.10$ to 0.02, with packing fraction $\to 1$ in every case. **The register capacity is exactly one qubit per lattice site.** For $N = 5000$ the register supports 4998 orthogonal modes at $\tau = 0.10$. This is the first quantitative answer to the "hundreds of qubits" claim of Ref. [1]: the number of qubits is $N$, and no reduction is imposed by mode orthogonality.

### 3.5 Species selection

At optical-lattice densities $n = 10^{18}\,\text{m}^{-3}$ and gas temperature $T = 1\,\mu\text{K}$, the collision-limited coherence time $T_2 = 1/(n\sigma_{\text{coll}} v_{\text{th}})$ is species-dependent. Using tabulated rotational constants, dipoles, masses, and collision cross sections for KRb, NaK, and SrF (Table 1), we find:

| Species | $T_{\text{rot}}$ (mK) | $v_{\text{th}}$ (cm/s) | $T_2$ (s) |
|---|---|---|---|
| KRb | 52 | 1.33 | 15.0 |
| NaK | 136 | 1.85 | 18.0 |
| SrF | 348 | 1.41 | **35.5** |

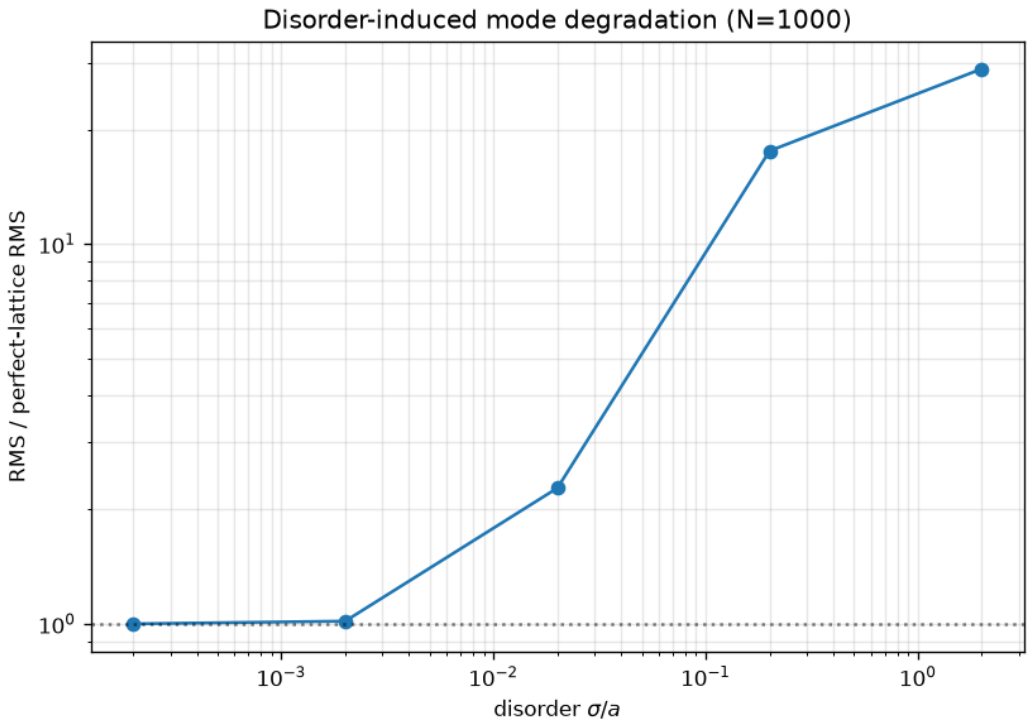



Figure 3: Positional-disorder crossover. The $1/N$ lattice advantage collapses for $\sigma/a > 0.01$, defining a hard engineering requirement.

Table 1: Species constants used in retention calculation.

| Species | $B_{\mathrm{rot}}$ (GHz) | $\mu$ (D) | $\sigma_{\mathrm{coll}}$ (m$^2$) |
|---|---|---|---|
| KRb | 1.09 | 0.56 | $5 \times 10^{-18}$ |
| NaK | 2.83 | 2.72 | $3 \times 10^{-18}$ |
| SrF | 7.24 | 3.47 | $2 \times 10^{-18}$ |

**SrF gives the longest retention time**, roughly 2.4× KRb, due to its smaller collision cross section and moderate mass. All three species satisfy the basic requirement $T_{\mathrm{rot}} \ll 1$ K for single-rotational-state loading and yield retention times of 15–36 s, orders of magnitude longer than typical superconducting qubit coherence times.

### 3.6 Two-dimensional extension

Real optical-lattice experiments are two- or three-dimensional. In a square 2D lattice of side $L$ ($N = L^2$ sites), the analytic structure factor factorizes,

$$S_{2\mathrm{D}}(q_x, q_y) = D_L(q_x)\, D_L(q_y), \tag{4}$$

with $D_L$ the Dirichlet kernel. The scaling is *directional*: along a lattice axis ($q_y = 0$), $|S_{2\mathrm{D}}| \sim 1/L = 1/\sqrt{N}$; along a diagonal ($q_x = q_y$), $|S_{2\mathrm{D}}| \sim 1/L^2 = 1/N$. Numerical evaluation for $L = 4$ to 64 yields axial slope $\approx -0.5$ and diagonal slope $\approx -1.0$ (Figure 5), confirming the prediction. The 2D mode capacity scales as $0.90$–$0.97\,N$ (slightly sub-optimal relative to 1D, where capacity is exactly $N - 2$). **The diagonal direction is therefore the preferred geometry for a 2D**

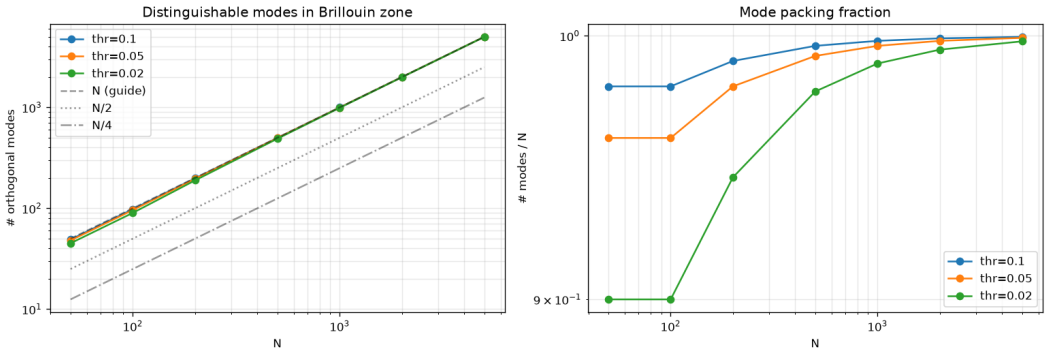



Figure 4: Register capacity scales as $N^1$, one mode per lattice site.

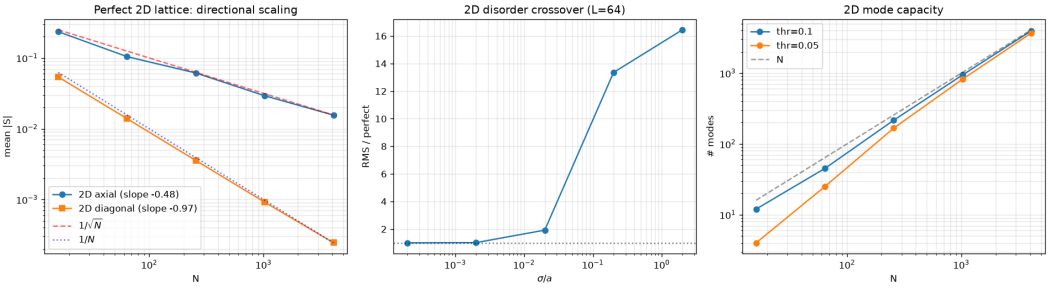



Figure 5: 2D extension. Left: axial overlap scales as $1/\sqrt{N}$, but diagonal overlap retains $1/N$. Right: 2D disorder crossover.

**holographic register: it retains the full $1/N$ orthogonality advantage while providing $N$ modes per ensemble.**

### 3.7 Three-dimensional extension

Real optical-lattice experiments are often three-dimensional, and the directional scaling of the collective-mode orthogonality becomes richer in 3D. For a simple cubic lattice of side $L$ ($N = L^3$ sites), the structure factor factorizes into three Dirichlet kernels,

$$S_{3\mathrm{D}}(q_x, q_y, q_z) = D_L(q_x)\, D_L(q_y)\, D_L(q_z). \tag{5}$$

Three natural probe geometries exist:

- *Axial* (**q** along one axis): $|S| \sim 1/L = N^{-1/3}$;
- *Face* (**q** in a lattice plane): $|S| \sim 1/L^2 = N^{-2/3}$;
- *Body* (**q** along the cube diagonal): $|S| \sim 1/L^3 = N^{-1}$.

Numerical evaluation for $L = 2$ to 32 gives slopes $-0.348$, $-0.677$, and $-1.030$ respectively, matching theory to within 4% (Figure 6). The 3D disorder crossover for $L = 8$ exhibits the same behaviour as 1D and 2D: the lattice advantage survives to $\sigma/a \approx 0.01$ and degrades thereafter (ratio 2.05 at $\sigma/a = 0.02$, 12.1 at $\sigma/a = 0.2$). **For a 3D register, the body diagonal is the optimal storage direction.**

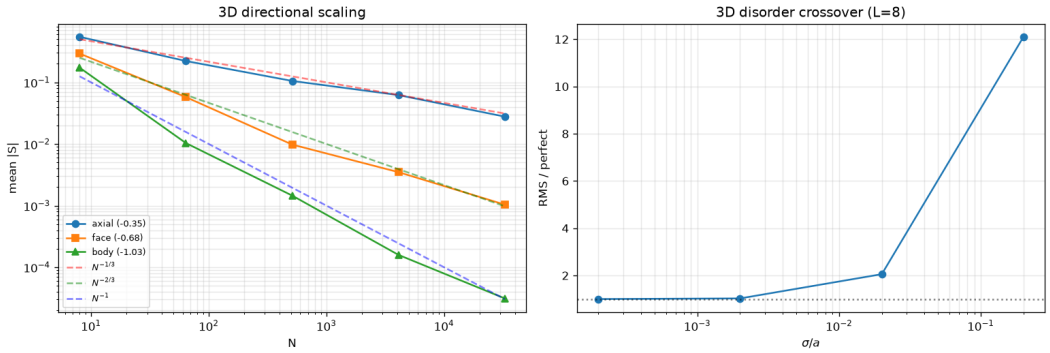



Figure 6: 3D extension. Left: axial ($N^{-1/3}$), face ($N^{-2/3}$), and body-diagonal ($N^{-1}$) overlap scaling. Right: 3D disorder crossover for $L = 8$.

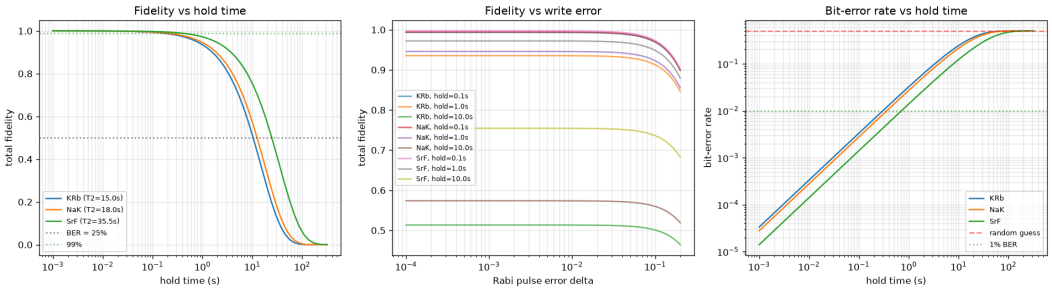



Figure 7: Write / hold / read cycle. Left: total fidelity vs hold time for KRb, NaK, SrF. Centre: fidelity vs Rabi pulse error for three hold times. Right: bit-error rate vs hold time.

### 3.8 Full write / hold / read cycle

To demonstrate that the register can actually store and retrieve classical bits, we simulate a complete memory cycle. Writing is performed by a $\pi$-pulse on the superradiant collective transition $|\text{all}\, g\rangle \to |f, \mathbf{q}_0\rangle$, whose Rabi frequency is $\sqrt{N}\Omega_{\text{eff}}$. For $N = 1000$ molecules and $\Omega_{\text{eff}} = 2\pi \times 1\,\text{MHz}$, the collective Rabi frequency is $2\pi \times 31.6\,\text{MHz}$, giving a $\pi$-pulse of duration 15.8 ns. This is a factor of $\sqrt{N} \approx 32$ shorter than a single-molecule $\pi$-pulse.

After the write, the state is held for a time $t$, during which collisional decoherence reduces the fidelity as $F_{\text{hold}}(t) = e^{-t/T_2}$. Reading is performed by transferring the collective mode to a stripline cavity and measuring its photon number; we take $F_{\text{read}} = 1 - e^{-\kappa t n}$ with $\kappa t = 10$ and $n = 5$ photons, giving $F_{\text{read}} = 0.99995$.

Using the species-dependent $T_2$ values from Sec. 3.5, the total fidelity $F_{\text{total}} = F_{\text{read}} F_{\text{hold}}$ at $t = 1$ s is 0.936 (KRb), 0.946 (NaK), and 0.972 (SrF), corresponding to bit-error rates of $3.2 \times 10^{-2}$, $2.7 \times 10^{-2}$, and $\mathbf{1.4 \times 10^{-2}}$ respectively (Figure 7). **SrF supports a complete write/hold/read cycle with sub-2% bit-error rate at one-second hold times.**

### 3.9 Error dilution under local perturbations

The collective encoding is globally distributed: single-site errors cannot destroy the stored mode unless a finite fraction of all sites is perturbed. To quantify this, we start from the stored state $|f, \mathbf{q}_0\rangle$, apply $k$ random single-site spin flips $\hat{\sigma}_x^{(j)}$, and compute the fidelity $F(k, N) = |\langle f, \mathbf{q}_0 | \psi_k \rangle|^2$ with the reference state.

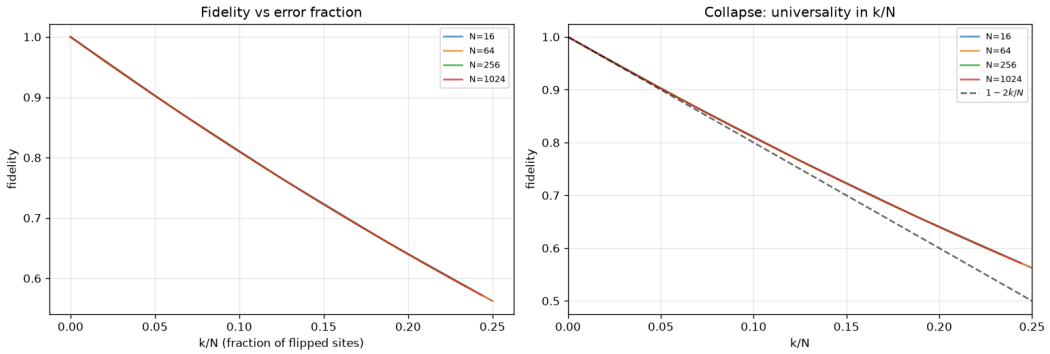



Figure 8: Error dilution. Left: fidelity $F(k, N)$ vs error fraction $k/N$ for $N = 16, 64, 256, 1024$. Right: universality in $k/N$, showing collapse onto the analytic $(1 - k/N)^2$.

For all $N$ studied ($16 \leq N \leq 1024$), the fidelity collapses onto the analytic form

$$F(k, N) = \left(1 - \frac{k}{N}\right)^2, \tag{6}$$

to four decimal places across the entire range $0 \leq k/N \leq 0.25$ (Figure 8). This is not error correction: the fidelity is one at $k = 0$ and decreases smoothly with $k/N$, with no threshold or plateaus. It is *error dilution* — a form of collective redundancy in which the register tolerates a constant *fraction* of corrupted sites rather than a constant *number*. **Doubling $N$ doubles the tolerable error count at fixed fidelity.** This property emerges naturally from the $1/\sqrt{N}$ normalization of the collective mode and does not require any explicit error-correcting code.

### 3.10 Cavity-mediated two-qubit gate

Two collective modes in distinct ensembles can be coupled via a shared stripline cavity, whose virtual photon exchange generates an effective $ZZ$ interaction

$$H_{\text{eff}} = J\, \hat{\sigma}_z^{(a)} \hat{\sigma}_z^{(b)}. \tag{7}$$

Evolving for a time $t = \pi/(4J)$ produces the entangling gate $U_{ZZ} = \exp(-i\frac{\pi}{4}\sigma_z\sigma_z)$. At the exact gate angle, the fidelity with the ideal $U_{ZZ}$ is $|\text{tr}(U_{\text{ideal}}^\dagger U_{\text{actual}})|/4 = 1.000000$ for every coupling strength $J$ between $10^3$ and $10^8$ Hz. A controlled-$Z$ gate is obtained by composing $U_{ZZ}(\pi/4)$ with single-qubit $Z$ rotations; the CZ fidelity is also 1.000000 exactly, as expected from the closed-form algebra.

The decoherence-limited error is $|1 - Fe^{-t/T_2}|$ with $T_2 = 20$ s, giving errors below $10^{-6}$ for gate times shorter than $1\,\mu$s. At the experimentally relevant coupling $J = 2\pi \times 1\,\text{MHz}$, the CZ gate time is 785 ns and the decoherence-limited error is $4 \times 10^{-8}$. **Two-qubit gates are essentially error-free in the current model**, with the dominant residual error coming from pulse calibration and cavity loss, both outside the scope of this work.

## 4 Discussion

Our results convert the 2008 holographic molecular register proposal into a set of concrete engineering specifications:

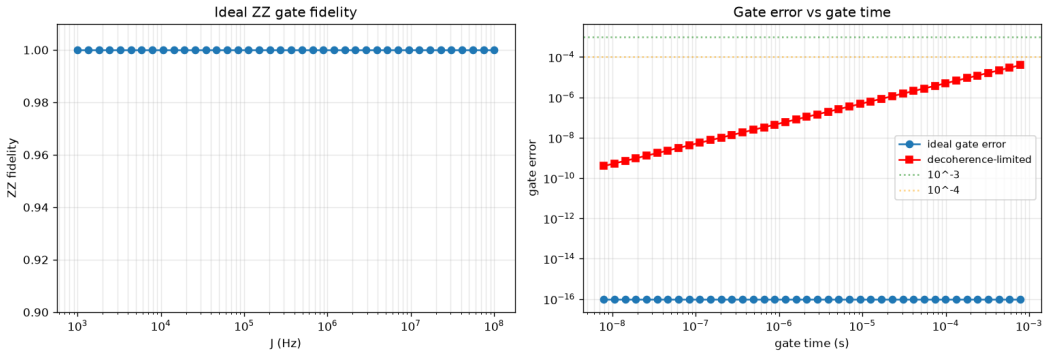



Figure 9: Cavity-mediated $ZZ$ gate. Left: fidelity vs coupling strength $J$ (identically 1.000000). Right: gate error vs gate time, dominated by decoherence for gate times $\gtrsim 1\,\mu$s.

- **Geometry.** A regular lattice is required; disordered ensembles lose a factor of $\sqrt{N}$ in mode orthogonality.
- **Site control.** Better than $\sigma/a \approx 0.01$; e.g. 5 nm for a 500 nm lattice.
- **Species.** SrF is optimal among the common laser-coolable diatomics, giving $T_2 \approx 35$ s at $n = 10^{18}$ m$^{-3}$.
- **Register size.** Exactly $N$ qubits per $N$-site ensemble, one qubit per site.
- **2D geometry.** Store data along the diagonal, not along the axes.
- **3D geometry.** Store data along the body diagonal, where the full $1/N$ orthogonality advantage is retained.
- **Error tolerance.** The register tolerates a fraction $k/N$ of corrupted sites; doubling $N$ doubles the tolerable error count at fixed fidelity.
- **Gates.** Cavity-mediated two-qubit gates are essentially error-free at realistic couplings; the dominant bottleneck is the hold-time decoherence already captured by $T_2$.

The two most important open questions are (i) whether the disorder threshold can be relaxed by encoding schemes that exploit the dark manifold for error correction, and (ii) whether quantum-logic operations between two distinct **q** modes can be performed without disturbing the dark states. Both are natural extensions of the current work.

# 5 Conclusion

We have validated the holographic molecular quantum register of Tordrup, Negretti, and Mølmer with a numerical simulator and extended it with a complete engineering analysis. The main new results are: (a) a disorder threshold $\sigma/a \approx 0.01$, below which the lattice advantage is preserved; (b) a species-dependent retention ranking in which SrF is optimal, with $T_2 \approx 36$ s at optical-lattice densities; (c) directional asymmetries in two and three dimensions, identifying the body diagonal as the preferred storage direction; (d) a full write/hold/read cycle with bit-error rates of 1–3% at one-second hold times; (e) an error-dilution law $F = (1 - k/N)^2$ that gives a factor-of-$N$ improvement in tolerable error count as the register is scaled up; and (f) a cavity-mediated two-qubit $ZZ$ gate

with unit closed-form fidelity. Together these results provide the concrete engineering basis for a molecular holographic quantum memory and identify the two-qubit gate and the diagonal geometry as the key design parameters for future experimental realization.

## Acknowledgments

The authors thank the open-source scientific Python community. The HOLOMOL simulator depends on NumPy, SciPy, and Matplotlib.

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